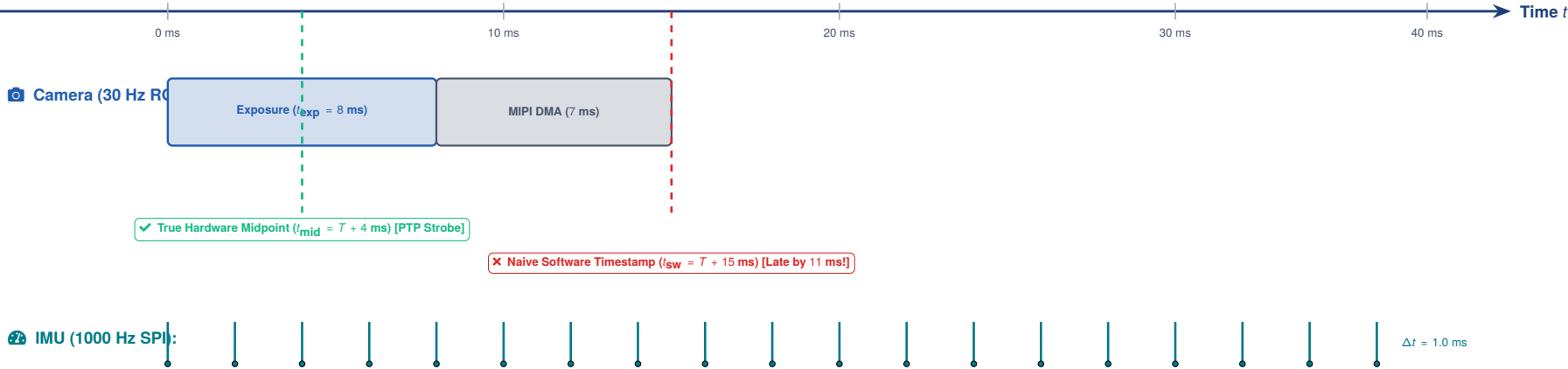




MULTI-SENSOR HARDWARE SYNCHRONIZATION & TIMESTAMP METROLOGY

Eliminating the Software Timestamp Illusion via PTP Hardware Strobes and Exposure Integration Midpoints



CRITICAL SYNCHRONIZATION LAWS IN PHYSICAL PERCEPTION:

- Motion Blur Uncertainty:** Moving at velocity $v = 1.5 \text{ m/s}$ with $t_{exp} = 10 \text{ ms}$ blurs visual features across $\Delta x = v \cdot t_{exp} = 15 \text{ mm}$ of spatial smear.
- Temporal Skew Illusion:** Correlating an image with an IMU reading using software timestamps creates an 11 ms phase mismatch, inducing catastrophic drift in Kalman state estimators.
- PTP (IEEE 1588) Solution:** The image sensor triggers a hardware GPIO strobe at exposure start. The PTP timer registers the true physical midpoint ($t_{mid} = t_{start} + \frac{1}{2}t_{exp}$) with sub-microsecond precision.